Automated Terms and Condition Analyser using Natural Language Processing

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***Abstract* — The widespread disregard for website terms and conditions poses a significant threat to personal privacy and security in today's fast-paced world. To combat this problem, we propose a system that utilizes natural language processing to analyse website policies and agreements. Our system starts by identifying terms and conditions pages through web scraping with the Beautiful Soup library[1]. Then, it employs NLP techniques such as Tokenization, Stemming, and keyword extraction to summarize and categorize content into three risk levels: high, medium, and low[2]. A user interface presents these categories and concise terms and conditions, enabling users to make informed decisions about acceptance. A crucial component of our system is the TextRank algorithm, which streamlines lengthy legal documents, making them more accessible and efficient while retaining essential information[3]. Our approach not only advances NLP but also underscores the importance of transparent, user-friendly online policies.**

Keywords — Natural language processing, Website terms and conditions, Web scraping, Beautiful Soup library, Stemming, Data protection, Tokenization, TextRank algorithm, Security.

I. INTRODUCTION

Our ability to obtain information, communicate with others, and conduct business has all been changed by the internet. However, a serious threat to one's security and privacy has also been brought about by the digital age. Users of websites are frequently required to accept extensive terms and conditions before using their services.

As a result, many users dismiss them and agree without fully comprehending the implications. This disregard for terms and conditions can lead to data leaks, identity theft, and other security breaches. Our technology is designed to enhance data security while simplifying the process for users to study and understand complex agreements. By identifying and removing Unnecessary content while highlighting essential keywords, our solution uses NLP to condense lengthy terms and conditions. This enables users to gain a better understanding of complex agreements and make informed decisions. By categorizing, summarizing, and clarifying website terms and conditions, our system empowers users to make informed decisions about accepting these agreements, ultimately strengthening their privacy and security in the digital age.

II. RELATED WORKS

Previous studies have explored comparable methods for identifying terms and conditions pages through web scraping. According to the presence of particular HTML tags and textual features, Hafiz et al. (2018) employed machine learning algorithms to identify terms and conditions websites[4]. Another study by Kim et al. (2019) suggested a method for locating terms and conditions pages based on the visual components and website layout analysis[5]. Zhang et al. (2018) adopted a novel approach to detect terms and conditions pages on Chinese e-commerce websites by combining rule-based and machine learning techniques[6]. A method to recognize terms and conditions pages was put forth by Wu et al. in their study from 2020, which was based on the examination of user behavior[7]. By leveraging web scraping and natural language processing to retrieve the terms and conditions page, text summarization is used to compress the text and identify keywords.

By Rada Mihalcea and Paul Tarau, "TextRank: Bringing Order into Texts"[8]. This study introduces TextRank, an algorithm for text summarization based on an unsupervised graph that ranks sentences according to their significance in the original text. Romain Paulus, Caiming Xiong, and Richard Socher's "A Deep Reinforced Model for Abstractive Summarization"[9].

The deep reinforcement learning model for abstractive text summarization that is suggested in this paper can provide coherent and useful summaries. The goal of text summarizing is to automatically produce a condensed version of a given text that still contains the essential details. Text summary can be a helpful tool for obtaining crucial information and insights from lengthy legal papers while reviewing terms and conditions.

Text summarization methods have been the subject of extensive research, including both extractive and abstractive methods. Whereas abstractive approaches create new phrases that accurately reflect the substance of the original text, extractive methods choose a subset of sentences from the

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original text to create the summary.

III. PROPOSED METHODOLOGY

A. Problem Definition

In today's digital landscape, complex website terms and conditions pose a serious threat to personal privacy and security. Users often overlook these agreements, leading to potential security breaches. Existing research lacks a comprehensive solution. This study introduces an innovative system that combines web scraping and advanced natural language processing (NLP) to extract and analyze website content.

It categorizes terms and conditions by their impact on user rights, offering a clear understanding. The key goals include accurate content extraction, NLP preprocessing, risk categorization, and a user-friendly interface. This research not only addresses a critical privacy issue but also presents a groundbreaking solution, marking a significant advancement in the field of online security and user interactions with website policies.

In our methodology, we utilize web scraping techniques with the Beautiful Soup library to extract text from webpages, with a specific focus on identifying terms and conditions pages. This involves sending HTTP requests to various URLs and parsing the HTML content. Once the terms and conditions pages are identified through specific keywords, we employ Natural Language Processing (NLP) techniques. We start with tokenization to break down text into words and sentences for effective analysis. Stop word removal eliminates common, low-information words, while stemming ensures consistency. Keyword extraction is used to identify important terms.

The extracted keywords are then employed in risk categorization, which assesses the potential impact of each clause on user rights, resulting in high, medium, or low-risk categories. We develop a user-friendly interface for presenting categorized terms and conditions, enabling users to make informed decisions regarding acceptance or rejection. Rigorous testing is performed to validate the system's effectiveness. Ethical guidelines are followed throughout the process to respect privacy and rights.

B. System Architecture

Figure 1 displays the system's working diagram for locating the terms and conditions webpage on the website. You can use a combination of web scraping and natural language processing techniques to check if a given web URL or web page is a terms and conditions page on a website when the user comes across the terms and conditions webpage.

To extract the text from a web page, use a web scraping package like BeautifulSoup. Remove all HTML tags, punctuation, and stop-words before pre-processing the text (common words that are unlikely to be useful for identifying the page as a terms and conditions page).

Apply text categorization to the pre-processed text using a natural language processing library like NLTK or spaCy. One method is to classify the text as a terms and conditions page or not using a supervised machine learning technique like Naive Bayes or Support Vector Machines. You'll need a dataset of web pages that have been classified as terms and conditions pages or not in order to train the classifier. This dataset can be produced by manually selecting and tagging a sample of web pages from different websites.

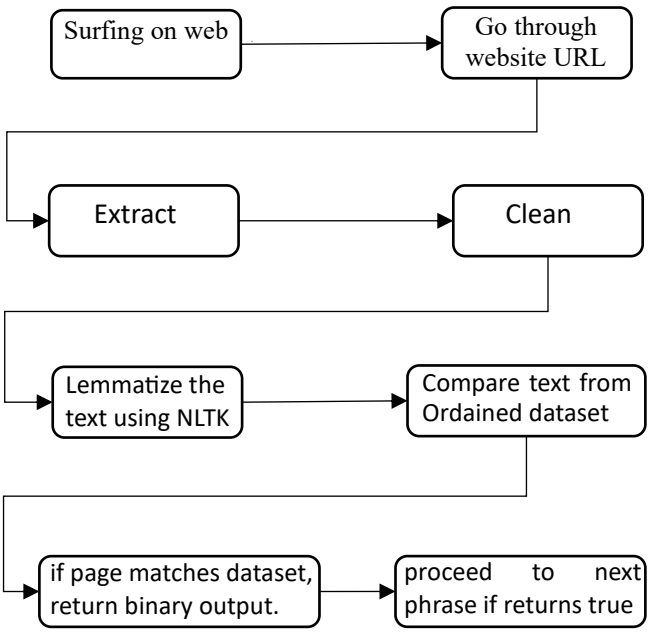

Fig. 1. Working diagram of the proposed system of T&C page detection.

The algorithm for determining if a given URL is a terms and conditions page or not, includes numerous processes, such as pre-processing, feature extraction, training data, model selection, model evaluation, and prediction. During the pre-processing stage, the text content of the webpage is extracted using the BeautifulSoup package. Using NLTK, the HTML tags, punctuation, and stop-words are eliminated from the text. The words in the text are additionally lemmatized using NLTK to return them to their original forms. In the feature extraction stage, significant features are gleaned from the pre-processed text, such as the existence of terms and conditions, legal disclaimers, privacy policies, etc.-related keywords. The text content is represented as a numerical feature vector using methods like bag-of-words or term frequency-inverse document frequency (TF-IDF).

A machine learning algorithm, such as Naive Bayes, is used to categorise the webpages in the model selection step. The chosen model is trained using the training dataset, and its performance is assessed using the validation dataset using the necessary metrics, including recall, recall time, recall accuracy, and F1-score. To improve the performance of the model, hyperparameter tweaking approaches like grid search and random search are used.

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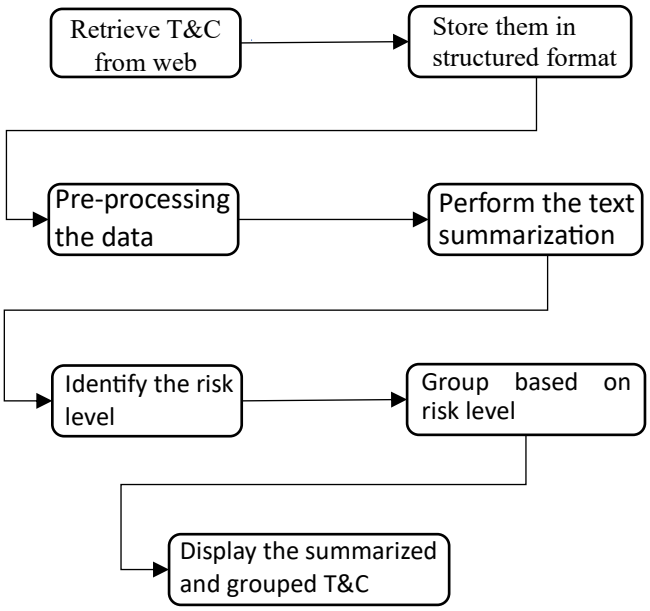

Fig. 2. Working diagram of the proposed system of summarizing and grouping the T&C

The trained and improved model's performance is assessed using a test dataset of unexplored webpages in the model evaluation step. In the test dataset, the model's performance metrics like accuracy, precision, recall, and F1-score are calculated.

The text content of a new webpage is pre-processed and its features are retrieved using the same methods as in the previous steps in the prediction step. In order to determine whether or not the webpage is a terms and conditions webpage, the feature vector is then supplied into the trained model.

By using these procedures, terms and conditions webpages can be correctly identified, making it easier for people to use the internet.

Following the discovery of the terms and conditions page using NLTK and beautifulSoup, we go on to summarise and organise the terms and conditions. The risk level of each phrase that has been condensed is determined after the terms and conditions have been summarised. A variety of methods, including keyword matching, rule-based approaches, and machine learning, can be used to do this. For instance, terms and conditions that can endanger consumers can be found using a predefined list of dangerous keywords and phrases. Alternately, the terms and conditions can be categorised based on their content using machine learning methods like Scikit-learn or TensorFlow.

The final step is to group the condensed terms and conditions according to how risky they are. This work can be accomplished using clustering methods like hierarchical or k-means clustering. These algorithms classify similar terms and conditions according to their content and level of risk, making it simpler to spot groups of problematic terms and conditions. The generated summaries and groups can be shown on the website in a user-friendly fashion, like a table or list, with the risk rating and summary given for each term and condition.

C. Implementation

Each component of the project is broken down into smaller tasks, and they are all listed below.

URL extraction and re-hash the data: For performing this URL extraction, we use beatifulSoup. If the request is successful, a string representing the webpage's HTML is retrieved. The HTML parser from BeautifulSoup is then used to read the HTML text[10,15]. We may explore the HTML tree structure and extract particular elements, like as paragraphs, headings, or lists, that contain the text content we are interested in using the BeautifulSoup object. The BeautifulSoup object's text attribute, which returns the textual content of an element and its descendants, can be used to extract the actual text from these elements.

BeautifulSoup: Import Beautifulsoup and requests library. Send a GET request to the web page you want to scrape Create a BeautifulSoup object by parsing the HTML content of the response. Use BeautifulSoup's various methods to extract the text content you want. For example, you can use the get_text() method to extract all the text on the page[16,19]. You can also use BeautifulSoup's various selectors to extract specific elements or attributes from the page. For example, you can use the find_all() method to find all the links on the page and extract their href attributes. Overall, BeautifulSoup is a powerful and flexible library that allows you to extract almost any information you want from a web page. However, it's important to keep in mind that web scraping can be illegal or unethical in certain circumstances, so make sure to consult the website's terms of service and use the data responsibly.

Lemmatization: First, you need to install the nltk library by running the 'pip install nltk' command in your terminal or command prompt. Import the nltk and necessary libraries. Download the necessary nltk resources 'punkt' and 'wordnet'. Create a 'WordNetLemmatizer' object[11,14]

Tokenize the text into individual words Lemmatize each word using the 'lemmatize()' method Lemmatization is a powerful technique for reducing words to their base or root form, which can help improve the accuracy and efficiency of natural language processing tasks such as text classification and information retrieval.

Compare text from pre-defined dataset: From the scikit-learn library, we import pandas, CountVectorizer, MultinomialNB, accuracyscore, precisionscore, and recall score. Bring up the terms and conditions. we read into a panda DataFrame the terms and conditions text from a CSV file along with their related labels. Create training and testing sets from the dataset: Using the pandas DataFrame sample technique, this part divides the dataset into training and testing sets at random. Using the CountVectorizer technique

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from Scikit-Learn, this portion extracts features from the terms and conditions text. Common English words like "the," "and," "a," and "a" are removed with the stop-words="English" option. Naive Bayes classifier training: we apply the fit method to the training set to train a Multinomial Naive Bayes classifier. Model testing: In this part, the trained model is put to the test on the testing set by utilising the predict technique to predict the labels of the test set. Model evaluation: The accuracy, precision, and recall metrics used in this section are used to assess the model's performance. At the conclusion of the code, the metrics are written to the console.

Terms and conditions summarization: Text summary is the practise of condensing a text to give the user a high-level overview of the entire text while saving them time and effort. The Natural Language Toolkit (NLTK) package is a popular option for text summarization, which is a function of natural language processing (NLP)[12]. The NLTK module gives computers the ability to comprehend natural language, whether it be spoken or written and then translated into numbers. The steps below are used to conduct text summarization Eliminating stop-words, unnecessary letters, and performing stemming or lemmatization during text pre-processing use Genism or TextRank or other similar libraries to implement extractive or abstractive text summarising techniques demonstrated in Fig.2[17].

Using methods like keyword matching or machine learning algorithms to determine the significance and relevance of the condensed text, A website's terms and conditions can be better understood and communicated by grouping the summarised content according to its relevance and importance.

a. **Creating sentences from paragraphs**: When creating sentences from a paragraph, start where the full stop occurs. This is crucial for the subsequent data cleansing step.

b. **Text cleaning**: The text must be cleaned after paragraphs have been broken up into sentences. Any special characters, digits, and stop words must be removed from the data in order to tidy it up.

c. **Tokenization**: To tokenize is to divide phrases down into individual words. Later, these words are obtained as an array with commas between each word.

Textrank: A particular use of the PageRank algorithm is the Textrank algorithm. The algorithm's first step is to tidy up the text. This is due to the text's frequent noise and inconsistencies. To exclude all stop words, character filters can be used. The next step after cleaning the text is to separate it into tokens or potential keywords. On the tokens that wouldn't make effective keywords after the words have been broken up into smaller units called tokens. Part-of-speech tagging is applied in this case. Sentences are broken up into words for part-of-speech tagging. These words fall within the noun, pronoun, adverb, and verb categories.

Working Procedure: The interface will also provide the proportion of three accessible categories for quicker examination.

Step 1: We use Beautiful Soup to parse the HTML text and submit a request to the terms and conditions page on the *Sample website* (www.samplewebsite.com/t&c) using the requests library.

Step 2: Next, we utilise BeautifulSoup to look for links that contain the words "*terms*" or "*conditions*" in order to identify the link to the Terms and Conditions page.

Step 3: The terms link variable will include a link to the Terms and Conditions page on the *Sample website*, and the programme will then move on to the following step. If website return Boolean value as true then proceed for further processing

"Acceptance of Terms: You accept these terms and conditions by visiting or using our website.

Intellectual Property: The material, pictures, graphics, and logos on this website are all the owner's and are covered by copyright laws.

Website Use: You may only use this website in line with these terms and conditions and for reasons that are permitted by law.

Disclaimer: The material on this website is only provided for general informational purposes and is not intended to be advice on any professional, financial, or other problems. We make no representations or warranties of any kind, express or implied, as to the completeness, accuracy, reliability, suitability, or availability for any purpose of the website or related graphics given on the website.

Limitation of Liability: We expressly disclaim all liability for any direct, indirect, incidental, special, or consequential damages that may arise from the use of or the inability to use this website or the information, goods, services, or associated visuals provided on the website.

Links to Other Websites: We have no control over the links that may appear on our website leading to other websites. Any connected website's practises, privacy policies, or content are not our responsibility.

Privacy Statement: We take seriously our responsibility to secure your personal information.

Governing Law: Without regard to any considerations of conflict of law, these terms and conditions shall be governed by and construed in accordance with the laws of [State/Country].

Terms and Conditions Changes: We have the right to make changes to these terms and conditions at any time and without prior notice".

basis of whitespace and punctuation, words are divided into tokens. Now we divide the above paragraphs into condensed sentences.

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Step 4: To obtain the webpage's HTML code, one uses the "requests library. The text content of the webpage is then obtained using the "response.text" element.

Step 5: Based on the level of danger, the text's content is categorised into three groups. The NLP strategies applied for summarising include:

a) Tokenization of Sentences: The material is broken up into separate sentences. Stop-word Removal: Frequent stop words, such "and," "the," and "a," are eliminated from the sentences[20].

b) Stemming: Words are stripped of their suffixes to return to their original form, so the words "modify," "modifies," and "modified" are all considered to be one word[21,22].

c) Keyword Extraction: Sentences are divided into categories based on whether they include high-, medium-, or low-risk-related keywords.

The parts of the terms and conditions are taken into account using the following predefined dataset:

Datasets:[Acceptance of Terms ,Intellectual Property ,Use of Website ,Disclaimer ,Limitation of Liability ,Links to Third-Party Websites ,Privacy Policy , Governing Law ,Changes to Terms and Conditions Privacy, Policy, Prohibited Uses Promotions, Proprietary Rights Quality Refund, Registration Remedies Representations, Restrictions Severability Sharing of Information, Suspension ,Taxes Termination, Third-Party Services, Trademarks Transparency Unsolicited Ideas, User-Generated Content, Use of Website Validity, Verification Viruses Waiver]

Step 6: The summarization_text() function is a critical component of the proposed methodology for analysing website terms and conditions. This function is responsible for applying Natural Language Processing (NLP) techniques to condense the lengthy terms and conditions into a more manageable and understandable form. Here's a breakdown of what the summarization_text() function will do:

TextRank is a method for summarizing text by treating it as a network of connected sentences. It's similar to how Google ranks web pages. In a text, sentences are like web pages, and the connections between them are like links. If a sentence is connected to many other sentences, it's considered important. TextRank analyses these connections to figure out which sentences are most significant.

LSA is a technique that uses math to understand the main topics in a large piece of text. It looks at patterns in the way words are used to find the hidden meanings or themes. In summarization, LSA helps identify sentences that are most representative of these themes.

High Risk: Phrases that refer to amending or altering the Terms and Conditions since this might materially affect the rights of the user.

Medium Risk: Sentences that mention terms associated with intellectual property rights may have a minimal effect on the rights of the user.

Low Risk: Phrases that don't contain any terms with a high or medium risk.

Calculation:

- Let T = {t1, t2, …, tn} be a set of n terms and conditions documents.
- Let S = {s1, s2, …, sm} be a set of m risk-related keywords or phrases.
- For each term ti in T, compute a risk score ri as follows:

$$ri = \frac{sum(W_{ij} * S_j)}{n} \quad (1)$$

- where wij is the weight of keyword sj in term ti, and n is the total number of keywords in term ti as given in (1).
- Group the terms and conditions documents based on their risk scores. For example, one possible grouping could be:
- Low risk: terms with a risk score < 0.25
- Medium risk: terms with a risk score >= 0.25 and < 0.5
- High risk: terms with a risk score >= 0.

The formula for TextRank can be expressed as follows:

$$Score(Si) = (1 - d) + d * \sum_{j=1}^{n} \left(\frac{W_{ji}}{\sum_{k=1}^{L_j} W_{jk}} * Score(sk)\right) \quad (2)$$

where Si is the score of sentence i, d is a damping factor (usually set to 0.85), n is the total number of sentences in the document, wji is the weight of the edge between sentence i and j (which is based on the similarity between the two sentences), Lj is the number of outgoing links from sentence j, Sk is the set of sentences that link to sentence k, and Score(Sk) is the score of sentence k as given in (2). The algorithm iteratively calculates the scores of all sentences until convergence.

Step 7: A user interface that displays the categories and the abbreviated terms and conditions is used to present the results. The user has the option to accept or reject the terms and conditions based on the category[13]. Moreover, the interface displays the proportion of sentences in each of the three categories.

IV. RESULTS AND DISCUSSION

The project classifies a website's terms and conditions page using a combination of web scraping and NLP methods. The software first parses the HTML text and locates the link to the terms and conditions using the Beautiful Soup library. The requests library is then used to get the webpage's text content.

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After that, the algorithm categorises the text into low, medium, and high-risk levels using NLP techniques including tokenization, stemming, and keyword extraction. The example above was tested in an automated terms and condition analyser[18].

Output for the given sample:

Group: Very risky

- Disclaimer: Not intended as legal, financial or other professional advice.

Group: Risky

- Limitation of Liability: Not liable for any damages arising out of use or inability to use website.

- Links to Third-Party Websites: Not responsible for content, privacy policies or practices of linked websites.

Group: Not risky

- Acceptance of Terms: Agree to be bound by these terms and conditions.

- Intellectual Property: Content on website is property of website owner.

- Use of Website: Use website only for lawful purposes and in accordance with these terms and conditions.

- Privacy Policy: Committed to protecting personal information.

- Governing Law: Terms and conditions governed by laws of [State/Country].

- Changes to Terms and Conditions: Reserve right to modify terms and conditions without prior notice.

Very risky: 14.29%

Risky: 28.57%

Not risky: 57.14%

V. CONCLUSION

In this study, we suggested a web scraping tool that uses the Beautiful Soup library to extract text content from a webpage Address and detect the existence of terms and conditions. The text was subsequently condensed using Natural Language Processing (NLP) techniques such sentence tokenization, stop-word removal, stemming, and keyword extraction. On the basis of their importance and level of risk, we further divided the summarised terms and conditions into three groups: high, medium, and low.

The three categories and the related shortened terms and conditions are shown in a user interface we created. Depending on the category, the user can then agree or reject the terms and conditions. The interface also displays the proportion of terms and conditions in each of the three categories that are currently accessible. The suggested approach may help consumers make wise selections when accepting the terms and conditions of a website.

According to our research, many website terms and conditions might put consumers at danger, especially in terms of privacy and legal responsibility. To further safeguard users' rights and lessen possible harm, we urge website owners and regulators to take action to increase the openness and clarity of these papers. Future studies in this field may examine how different interventions, including standardised templates or instructional materials, might enhance the standard and security of online terms and conditions.

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